



For Open World

This article reflects on how open source is the natural outcome of, and fits in so well with, the global software ecosystem, which includes the Internet and the World Wide Web.

I will begin by explaining what this article is not. It is not technical, nor is it a tutorial. It does not have lines and lines of code, nor is it a set of tips on how to get something done. While I have immense respect for writers of such articles, who help newbies like me learn new stuff, I am just not experienced enough to write like that yet.

But I am, nevertheless, an open source fanatic, a lover of all things Linux—and I have, over the past few years, developed a certain understanding of the bigger picture related to the whole concept of open source. And this is what the article is about.

Before we talk about open source, I'd like to generalise the concept of openness, by bringing into our discussion an *open system*. If you look up Wikipedia, you will find a host of search results, in the fields of computing, systems theory, control theory and a whole lot of others. None of them are the 'open system' that I'm referring to. However, there are similarities between my open system and all the open systems featured there.

So what is an open system then? In its bare essentials, it is a system that has no restrictions. It is open for all, to all. Anyone and everyone can enter the system and be a part of it; even work for it. This brings me to the next keyword of importance: *everyone*.

What sets apart an open system from one that is not open? That one word, *everyone*. It is the community of an open system that makes it different from a closed system. If an open system survives, it is because of its community alone, without which an open system is virtually dysfunctional or useless. And what is this community? It is a group of people who are part of the system. They use the services of the system or help in developing it. This community is made up of two kinds of people: those who care about the system and

are willing to work for it, and those who trust the system and use its services. Collaboration within the community is what keeps an open system from extinction.

The finest example of such a system is Wikipedia. It isn't difficult to figure out who makes up this community—it is virtually the entire computer-savvy population of the human race. There are millions of people who use Wikipedia every single day—blindly using the information they access on it for their daily needs and activities. The millions who use Wikipedia form the user-faction of the community—those who trust it. Again, the same community also has hundreds and thousands of other people who actively maintain Wikipedia. These include the authors, editors, and donors, because of whom Wikipedia manages to exist... and grow.

This community of Wikipedia ensures that the dynamism of the system is maintained, that errors are removed, and that information is up to date. Thus, it is this community that helps Wikipedia to survive.

I now jump to the next keyword—*information*. Pointless as the question might seem, what is information? Simply put, it is an organised collection of facts and figures. The important thing about information is that it needs to be dynamic; otherwise, it is of no use. This is a crucial point that I wish to stress upon—that static information is almost as useful as a snooze button on a smoke alarm. It can even prove to be dangerous. Information needs to be dynamic. It needs to flow. And it should flow without hindrance, within a system. Any disruption of information dissemination can create imbalances within the system, and this can adversely affect the system. The reason for this will become clear after I elaborate on the next big topic here, which is self-learning systems.